

Siwon Lee

Email : lsw021228@g.skku.edu
Mobile : +82-10-4938-6605

Linkedin: <https://www.linkedin.com/in/siwon-lee-4a543520a/>
Site: <https://siwon-mathnote.kro.kr/Profilen>

I want to analyze monetary policy on the open market. More specifically, I want to explain international trade based on the central bank's monetary policy. Also, I have an interest in Applied econometrics using supervised learning.

EDUCATION

University of Washington in St.Louis

Ph.D. Economic

MO, USA

Aug 2024

- Anticipated Graduation: Aug 2029

Sungkyunkwan University

B.A. in Global Economics, B.Math Mathematics

Seoul, Korea

Mar 2021-Aug 2024

- Cumulative GPA: 4.17/4.5, Global Economics: 4.31/4.5, Mathematics: 3.94/4.5
- Math: Analysis, Basic Mathematics for AI, Differential Geometry, Probability Theory, Computer Applied Mathematics
- Software: Operating System, Intro to DB, Data Structures, Intro to AI, System Programming, Algorithm
- Economics: Adv Macroeconomic Theory, Game Theory, Econometrics, Mathematical Economics, Intro Financial Accounting

LUSEM & HANKEN University

Joint Winter Exchange School

Sweden and Finland

Jan 2023-Jan 2023

- Visited KPMG corporation in Helsinki and discussed sustainability.
- Discussed critical thinking and examples of sustainability in each country.

WORK EXPERIENCE

TRC Research Assitant

Research Assistant

SKKU, Korea

Nov 2023-Feb 2024

- Organized from 111 pdf reports written in natural language to Excel.
- Fixed imperfect or mismatched data and recorded outlier cases.

Wisdomagora

Editor Management Intern

Seoul, Korea

Aug 2022 - Feb 2024

- Organized online article uploaded the schedule and checked image copyrights.
- Edited online article content for the business and economic sector written by other countries' international high school students.

Department of College TA

Teaching Assistant

SKKU, Korea

Feb 2023 - Jun 2023

- Graded students code and received questions related to the course.
- Feedback student's regular and extra studies, and suggest new ideas.
- Feedback student's team project related to their idea and implementation.

SKKU International Summer Semester TA

Teaching Assistant

SKKU, Korea

Jun 2022 - Jul 2022

- Communicated between professor and director related to lecture course or attendance problem
- Supported culture event as orientation or graduation targeted to exchanged student

RESEARCH

Analyze Impact of Monetary Policy on Construction Investment and Facility Investment

Coworker

SKKU, Korea

Mar 2024 Ongoing

- Analyze impact of monetary policy on investment using three paths as asset price, exchange rate, and interest rate.
- Analyzed South Korea case such as work-out of TaeYeong Company and LegoLand.
- Using Bayesian VAR and error correction model, statistical prove our statement.

Analyze effect of Monetary Policy to Technological Progress: NACIS334(Korean)

Team Leader

DB, Korea

Dec 2023 - Feb 2024

- Specializing in analyzing monetary policy effects on industries, exemplified by NACIS334 sector research.

- Applying statistical methods such as constrained variable analysis, latent variable analysis, regression, and TVP-FAVAR.
- Formulated theoretical hypotheses mathematically and empirically validated them using data.

Global Economic Individual Research 1(English)

Team Member

SKKU, Korea
Jun 2022 - Jul 2022

- Solving solow model problem using numerical method and python numpy library
- Researched solow model with changing several model's assumption such as s is constant
- Analyzed United States economic data using FRED API and python

Global Economic Individual Research 2(Korean)

Team Member

SKKU, Korea
Feb 2023 - Jun 2023

- Analyzed similarity between 2008 financial crisis and currently economic intension related to pandemic and war.
- Using TVP-FAVAR model, analyze FCI and macro economic index based on "The Term Structure of Growth-at-Risk"
- Analyzed using statistical model of TVP-FAVAR model changed variables using python.

Effect of Internet Reviews on the Demand of Experience Goods: Korean Movie Industry(English) SKKU, Korea

Team Member

Aug 2023 Ongoing

- Crawling review data from the Daum site and standardizing natural language and scoring using Python
- Implementing basic demand function of movie, and determine to situation.

Calculate Derivates and Options Expected Value(Korean)

Team Leader

SKKU, Korea
Sep 2022 - Dec 2022

- Understand basic financial markets and modeling, in particular, black-scholers model and binary price determinant model.
- Construct new price forecasting model using LSTM, and multiple ANN method.
- Analyze and select significant economic variable using classical macro models and algorithms.

Forecasting Inflation and Economic Recession: Comparison of Econometrics and Machine Learning(English)

Aug 2022

College of Economics

- Analyzed and forecasting inflation and GDP using ANN and classical econometrics method
- Reviewed progress and output of ANN model and econometrics model

Forecasting Changing Phase of Financial Market When Implementation CBDC

Feb 2022(Korean)

DB financial investment

- Researched property of bonds, stocks, and currency swap future
- Calculated value and risk of financial products using value evaluation equation

Implementation of CBDC to Reduce Public Antipathy of Privacy Violations

Aug 2021(Korean)

College of Economics

- Analyzed public's dislike about private invasion any CBDC type and monetary policy

TEACHING EXPERIENCE

Young Economic Education Supporters

Teacher

Bank of Korea, Korea
Apr 2023 - Ongoing

- Lectured element school and middle student on role of BOK, currency and inflation, and history of currency.
- Lectured didactics and make material of lecture.

Sungkyun Tutoring

Mentor

SKKU, Korea
Mar 2022 - Ongoing

- Built note when used python language tutoring and designed programming problems
- Educated concept of list, loop, and function and thinking method of algorithm
- Coached student's programming code and revised or assisted their code

EXTRACURRICULAR ACTIVITIES

Becoming Economist

Vice President, Team Member

SKKU, Korea

Mar 2022 - Jun 2024

- Reviewed several economic articles related to estimated of schooling, Rouwenhorst Method, and hyperbolic discount function.
- Constructed English certification study as TOEFL and GRE.
- Prepared study abroad as doctor program with OB meeting.

Global Lealers in Finance(GLIF)

Team Member

Seoul, Korea

Mar 2021 - Mar 2022

- Summarized important economic indicators such as S&P 500, KOSPI, and WTI rates to last market trends
- Abstract and translate daily articles from *WSJ*, *FT*, and *BB* in the field of macroeconomic news
- Structured fictional customer's portfolio and Analyzed Mezzanine and non-list company in an Academic-affiliated program

MIMIC

Team Member

SKKU, Korea

Sep 2022 - Jun 2024

- Communicated mathematic thoerm such as four color theorem in regular seminar
- Studying regular lecture as binominal mathematic or linear algebra.
- Lectured Nash's existence theorem and minimax theorem of Von.

Seekers

President

SKKU, Korea

Mar 2022 - Jun 2024

- Debated economic and social issues as Middle Interest Rate, Global Supply Chain, and others.
- Participated mentoring with senior graduated of SKKU economic department.

PROJECT

Analzed relationship between COVID-19 and trade

Team Leader

SKKU, Korea

Apr 2022 - Jun 2022

- Collected and analyzed COVID-19 data and international trade data using python
- Learning OLS regression with several different equation

Learning-Fair

Colleague Depart. Dean's Awards

SKKU, Korea

Sep 2021 - Nov 2021

- Assembles firm's financial statement, stock price, and trading volume data then standardized to excel data
- Implemented mock investment and classify investor tendency program using python library of pandas, mpl-finance

Project of Convergence Basics

Team Member

SKKU, Korea

Jun 2021 - Aug 2021

- Designed new items, using design thinking process as lottos-blossom & prototype test
- Collaborated with co-workers and delivered in front of professors and the CEO

Calculated Production Possibility Function

Team Member

SKKU, Korea

Apr 2021 - Jun 2021

- Calculated production possibility function on two different products and labor
- Solved feasible problem when gives each products price and budget line

AWARDS

2022 SKKU Annual Conference on Economics

Top Prize

SKKU, Korea

Jun 2022 - Aug 2022

- Analyzed neuron network method for economic research and comparison forecast efficiency to classical econometrics model.
- Tested several neuron network and econometrics model using past 50 years U.S. statistical data.

2022 Capston Design and AI Hackerton

Encouragement Prize

The Korean Association of Computer Education

Oct.13.2022

- Structured business canvas about new business item based on financial market and AI.
- Implemented AI model and front web design using R, LSTM, and time-series data.

Sungkyun tutoring

Excellence Prize

SKKU, Korea

Aug 2022 - Dec 2022

Facebook Hacker Cup 2021

Round 2

Facebook, Online

Aug 2021 - Sep 2021

Analyzing of Big Data contest by NH investment

Participation Prize

NH investment

Nov 2022

2022-1 Dean's List

SKKU, KOREA

CONFERENCES ATTENDED

SKKU-KEIO Research Symposium	SKKU and KEIO, Korea and Japan
<i>Speaker</i>	<i>Jan 2024</i>
2024 Colloquium: Decoupling and Application	SKKU, Korea
<i>Participation</i>	<i>Apr 2024</i>
International Conference on Einstein and Harmonic Manifolds	SKKU and Samsung, Korea
<i>Participation</i>	<i>19-22 Jun 2023</i>
MULTILINEAR OPERATORS IN HARMONIC ANALYSIS	SKKU, Korea
<i>Participation</i>	<i>26 Oct 20223</i>
MATLAB Expo Korea	MATLAB, Korea
<i>Participation</i>	<i>Jun 2022</i>
2022 SKKU International Conference: Trends in Digital Economy and Finance	SKKU., Korea
<i>Participation</i>	<i>May 2022</i>

SCHOLARSHIP

Scholarship for Academic Excellence	Feb 2022
Scholarship for Academic Excellence	Aug 2022
Scholarship for Global Exchange Student	Mar 2023
Scholarship for Global Learning Leader	Dec 2023
Scholarship for Extracurricular Challenge	Jan 2024

ADDITIONAL INFORMATION

Achivements: TESAT Grade 2, ADsP, Certification of Cody Joint Event(MATLAB)	
Skills: Programming(C, Python, HTML, MYSQL, Latex), MS Office(PowerPoint, Excel), Mathematica, Premiere Rush	
Languages: Korean(Native), English(Intermediate)	
Interests: Artifical Intelligence, Machine Learning, Financial Market, Economics Recession, Heterogeneous-Agent	
Hobby: Table Tennis, Tennis, Listening Music, Watching NBA	
Strengths: Enthusiasm, Curiosity, Punctual, Trustworthiness, focused	